

Project title:

A chatbot-like system to teach writing with the support of a SLM

Project outline:

(192 words; allowed 100-200 words)

There is a held belief that technical writing is less important in CS, despite its importance in the industry and impact on popular skills like vibe-coding. Students may rely on AI for writing during their undergraduate degree and not develop the writing skills necessary to communicate their work.

The goal of this capstone project is to develop a chatbot-like system that merges fixed prompts with customized writing support from a Small Language Model (SLM), to teach CS students writing skills and aid students while writing assignments provided by instructors. In addition, the instructor and SLM will customize the writing experience based on student expertise, interests, and writing struggles.

The project will involve:

- Forming a web-based interface with an instructor and student view. Instructors will input assignments, student interests and student writing challenges. Students will view a chatbot and iteratively complete assignments based on SLM and fixed prompts.
- Fine-tuning a SLM that co-writes with and provides personalized feedback for students based on their gauged writing skill and instructor customization.
- Creating a backend handles storing instructor input, combining instructor customization, fixed prompts, and SLM output for students, and collecting chat logs for research analysis.